3. DIFFERENTIAL OPERATORS IN DIFFERENT COORDINATES SYSTEMS

Coordinate system	Operators' Expressions
Cartesian coordinates	$grad(f) = \nabla f = \frac{\partial f}{\partial x}\hat{x} + \frac{\partial f}{\partial y}\hat{y} + \frac{\partial f}{\partial z}\hat{z}$ $\mathrm{div}(\vec{A}) = \nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$ $curl(\vec{A}) = \nabla \times \vec{A} = \begin{vmatrix} +\hat{x} & -\hat{y} & +\hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{x} - \left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z}\right)\hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{z}$ $lap(f) = \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ $lap(\vec{A}) = \nabla^2 \vec{A} = \nabla^2 A_x \hat{x} + \nabla^2 A_y \hat{y} + \nabla^2 A_z \hat{z} = \nabla(\nabla \cdot \vec{A}) - \nabla \times (\nabla \times \vec{A})$
Cylindrical coordinates	$\nabla f = \frac{\partial f}{\partial \rho}\hat{\rho} + \frac{\mathbf{1}}{\boldsymbol{\rho}}\frac{\partial f}{\partial \phi}\hat{\phi} + \frac{\partial f}{\partial z}\hat{z}$ $\nabla \cdot \vec{A} = \frac{\mathbf{1}}{\boldsymbol{\rho}}\frac{\partial}{\partial \rho}(\boldsymbol{\rho} A_\rho) + \frac{\mathbf{1}}{\boldsymbol{\rho}}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$ $\nabla \times \vec{A} = \frac{\mathbf{1}}{\boldsymbol{\rho}}\begin{vmatrix} +\hat{\rho} & -\boldsymbol{\rho}\hat{\phi} & +\hat{z} \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \boldsymbol{\rho} A_\phi & A_z \end{vmatrix}$ $= \frac{\mathbf{1}}{\boldsymbol{\rho}}\left[\left(\frac{\partial A_z}{\partial \phi} - \frac{\partial \boldsymbol{\rho} A_\phi}{\partial z}\right)\hat{\rho} - \boldsymbol{\rho}\left(\frac{\partial A_z}{\partial \rho} - \frac{\partial A_\rho}{\partial z}\right)\hat{\phi} + \left(\frac{\partial(\boldsymbol{\rho} A_\phi)}{\partial \rho} - \frac{\partial A_\rho)}{\partial \phi}\right)\hat{z}\right]$ $\nabla^2 f = \frac{\mathbf{1}}{\boldsymbol{\rho}}\frac{\partial}{\partial \rho}\left(\boldsymbol{\rho}\frac{\partial f}{\partial \rho}\right) + \frac{\mathbf{1}}{\boldsymbol{\rho^2}}\frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}$ $\nabla^2 \vec{A} = \left[\boldsymbol{\nabla^2 A_\rho} - \frac{A_\rho}{\rho^2} - \frac{2}{\rho^2}\frac{\partial A_\phi}{\partial \phi}\right]\hat{\rho} + \left[\boldsymbol{\nabla^2 A_\phi} - \frac{A_\phi}{\rho^2} + \frac{2}{\rho^2}\frac{\partial A_\rho}{\partial \phi}\right]\hat{\phi} + \boldsymbol{\nabla^2 A_z}\hat{z} = \nabla(\nabla \cdot \vec{A}) - \nabla \times (\nabla \times \vec{A})$
Spherical coordinates	$\nabla f = \frac{\partial f}{\partial r}\hat{r} + \frac{\partial f}{\partial \theta}\hat{\theta} + \frac{\mathbf{1}}{\boldsymbol{r}\,\mathbf{sin}\,\boldsymbol{\theta}}\frac{\partial f}{\partial \phi}\hat{\phi}$ $\nabla \cdot \vec{A} = \frac{\mathbf{1}}{\boldsymbol{r^2}}\frac{\partial}{\partial r}(\boldsymbol{r^2} A_r) + \frac{\mathbf{1}}{\boldsymbol{r}\,\mathbf{sin}\,\boldsymbol{\theta}}\frac{\partial}{\partial \theta}(\mathbf{sin}\,\boldsymbol{\theta}\, A_\theta) + \frac{\mathbf{1}}{\boldsymbol{r}\,\mathbf{sin}\,\boldsymbol{\theta}}\frac{\partial A_\phi}{\partial \phi}$

$$\nabla \times \vec{A} = \frac{\mathbf{1}}{\boldsymbol{r^2 \sin\theta}} \begin{vmatrix} +\hat{r} & -\boldsymbol{r}\hat{\theta} & +\boldsymbol{r sin\theta}\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & \boldsymbol{r}A_\theta & \boldsymbol{r \sin\theta}\, A_\phi \end{vmatrix}$$

$$= \frac{1}{\boldsymbol{r^2 \sin\theta}} \left[\left(\frac{\partial}{\partial \theta} (\boldsymbol{r \sin\theta}\, A_\phi) - \frac{\partial \boldsymbol{r} A_\theta}{\partial \phi} \right) \hat{r} - \boldsymbol{r} \left(\frac{\partial}{\partial r} (\boldsymbol{r \sin\theta}\, A_\phi) - \frac{\partial A_r}{\partial \phi} \right) \hat{\theta} + \boldsymbol{r \sin\theta} \left(\frac{\partial}{\partial r} (\boldsymbol{r} A_\theta) - \frac{\partial A_r}{\partial \theta} \right) \hat{\phi} \right]$$

$$\nabla^2 f = \frac{\mathbf{1}}{\boldsymbol{r^2}} \frac{\partial}{\partial r} \left(\boldsymbol{r^2} \frac{\partial f}{\partial r} \right) + \frac{\mathbf{1}}{\boldsymbol{r^2 \sin\theta}} \frac{\partial}{\partial \theta} \left(\boldsymbol{\sin\theta} \frac{\partial f}{\partial \theta} \right) + \frac{\mathbf{1}}{\boldsymbol{r^2 \sin^2\theta}} \frac{\partial^2 f}{\partial \phi^2}$$

$\nabla^2 \vec{A} = \nabla(\nabla \cdot \vec{A}) - \nabla \times (\nabla \times \vec{A})$
or
$\nabla^2 \vec{A} = \vec{V} = V_r \hat{r} + V_\theta \hat{\theta} + V_\phi \hat{\phi}$ such that

$$V_r = \left[\boldsymbol{\nabla^2 A_r} - \frac{2}{r^2} A_r - \frac{2}{r^2} \frac{\partial A_\theta}{\partial \theta} - \frac{2}{r^2} \cot\theta A_\theta - \frac{2}{r^2 \sin\theta} \frac{\partial A_\phi}{\partial \phi} \right]$$

$$V_\theta = \left[\boldsymbol{\nabla^2 A_\theta} + \frac{2}{r^2} \frac{\partial A_r}{\partial \theta} - \frac{1}{r^2 \sin^2\theta} A_\theta - \frac{2\cot\theta}{r^2 \sin\theta} \frac{\partial A_\phi}{\partial \phi} \right]$$

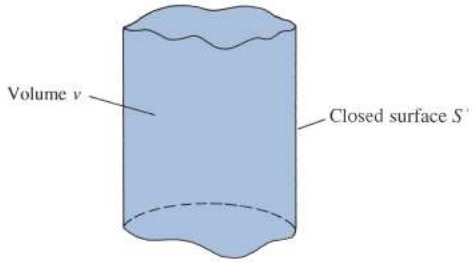
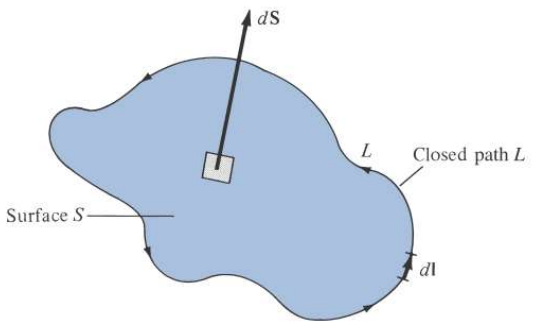
$$V_\phi = \left[\boldsymbol{\nabla^2 A_\phi} + \frac{2}{r^2 \sin\theta} \frac{\partial A_r}{\partial \phi} + \frac{2\cot\theta}{r^2 \sin\theta} \frac{\partial A_\theta}{\partial \phi} - \frac{1}{r^2 \sin^2\theta} A_\phi \right]$$

4. SOME VECTOR IDENTITIES

$\nabla(f + g) = \nabla f + \nabla g$
$\nabla(fg) = f\nabla g + g\nabla f$
$\nabla \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$
$\nabla \times (\vec{A} + \vec{B}) = \nabla \times \vec{A} + \nabla \times \vec{B}$
$\nabla \cdot (\nabla \times \vec{A}) = 0$
$\nabla \times (\nabla f) = 0$
$\nabla \cdot (f\vec{A}) = \vec{A} \cdot (\nabla f) + f(\nabla \cdot \vec{A})$
$\nabla \times (f\vec{A}) = (\nabla f) \times \vec{A} + f(\nabla \times \vec{A})$
$\nabla \cdot (\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \nabla)\vec{B} + (\vec{B} \cdot \nabla)\vec{A} + \vec{A} \times (\nabla \times \vec{B}) + \vec{B} \times (\nabla \times \vec{A})$
$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$
$\nabla \times (\vec{A} \times \vec{B}) = \vec{A}(\nabla \cdot \vec{B}) - \vec{B}(\nabla \times \vec{A}) + (\vec{B} \cdot \nabla)\vec{A} - (\vec{A} \cdot \nabla)\vec{B}$
$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$

5. DIVERGENCE THEOREM AND STOKES'S THEOREM

Line integral of $\vec{A}$ along an <u>open</u> curve L is $$\int_L \vec{A}.\overrightarrow{dl}$$ **Circulation** of $\vec{A}$ around a <u>closed</u> curve L is given by $$C = \oint_L \vec{A}.d\vec{l}$$	**Flux** of $\vec{A}$ on surface S is given by $$\psi = \int_S \vec{A}.\overrightarrow{dS}$$ **Total (net) outward Flux** of $\vec{A}$ on <u>closed</u> surface S (a volume shape) is given by $$\psi = \oint_S \vec{A}.d\vec{S}$$

Divergence theorem is given by $$\oint_S \vec{A}.d\vec{S} = \int_v \nabla.\vec{A}\,dv$$	Stokes's theorem is given by $$\oint_L \vec{A}.d\vec{l} = \int_S (\nabla \times \vec{A}).d\vec{S}$$
	

INDEFINITE INTEGRALS

If $U = U(x), V = V(x), a$ and b are constants, and C is an arbitrary constant,

$$\int a\,dx = ax + C$$

$$\int U\,dV = UV - \int V\,dU \text{ (integration by parts)}$$

$$\int U''\,dU = \frac{U''^{+1}}{n+1} + C,\ n \neq -1$$

$$\int \frac{dU}{U} = \ln U + C$$

$$\int a^U\,dU = \frac{a^U}{\ln a} + C,\ a > 0, a \neq 1$$

$$\int e^U\,dU = e^U + C$$

$$\int e^{ax}\,dx = \frac{1}{a}e^{ax} + C$$

$$\int x\,e^{ax}dx = \frac{e^{ax}}{a^2}(ax - 1) + C$$

$$\int x^2\,e^{ax}dx = \frac{e^{ax}}{a^3}(a^2x^2 - 2ax + 2) + C$$

$$\int \ln x\,dx = x\ln x - x + C$$

$$\int \sin a\,xdx = -\frac{1}{a}\cos a\,x + C$$

$$\int \cos a\,xdx = \frac{1}{a}\sin a\,x + C$$

$$\int \tan a\,xdx = \frac{1}{a}\ln\sec a\,x + C = -\frac{1}{a}\ln\cos a\,x + C$$

$$\int \sec a\,xdx = \frac{1}{a}\ln(\sec a\,x + \tan a\,x) + C$$

$$\int \sin n^2\,axdx = \frac{x}{2} - \frac{\sin 2\,ax}{4a} + C$$

$$\int \cos s^2\,axdx = \frac{x}{2} + \frac{\sin 2\,ax}{4a} + C$$

$$\int x\sin a\,xdx = \frac{1}{a^2}(\sin a\,x - ax\cos a\,x) + C$$

$$\int x\cos a\,xdx = \frac{1}{a^2}(\cos a\,x + ax\sin a\,x) + C$$

$$\int e^{ax}\sin b\,xdx = \frac{e^{ax}}{a^2 + b^2}(a\sin b\,x - b\cos b\,x) + C$$

$$\int e^{ax}\cos b\,xdx = \frac{e^{ax}}{a^2 + b^2}(a\cos b\,x + b\sin b\,x) + C$$

$$\int \sin a\,x\sin b\,xdx = \frac{\sin(a-b)x}{2(a-b)} - \frac{\sin(a+b)x}{2(a+b)} + C,\ a^2 \neq b^2$$

$$\int \sin a\,x\cos b\,xdx = -\frac{\cos(a-b)x}{2(a-b)} - \frac{\cos(a+b)x}{2(a+b)} + C,\ a^2 \neq b^2$$

$$\int \cos a\,x\cos b\,xdx = \frac{\sin(a-b)x}{2(a-b)} + \frac{\sin(a+b)x}{2(a+b)} + C,\ a^2 \neq b^2$$

$$\int \sinh a\,xdx = \frac{1}{a}\cosh a\,x + C$$

$$\int \cosh a\,xdx = \frac{1}{a}\sinh a\,x + C$$

$$\int \tanh a\,xdx = \frac{1}{a}\ln\cosh a\,x + C$$

$$\int \frac{dx}{ax+b} = \frac{1}{a}\ln|ax+b|$$

$$\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$$

$$\int \frac{xdx}{x^2+a^2} = \frac{1}{2}\ln(x^2+a^2) + C$$

$$\int \frac{x^2dx}{x^2+a^2} = x - a\tan\frac{x}{a} + C$$

$$\int \frac{dx}{x^2-a^2} = \begin{cases}\frac{1}{2a}\ln\frac{x-a}{x+a} + C \\ \frac{1}{2a}\ln\frac{a-x}{a+x} + C\end{cases} x^2 > a^2$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln\left(x + \sqrt{x^2 \pm a^2}\right) + C$$

$$\int \frac{xdx}{\sqrt{x^2+a^2}} = \sqrt{x^2+a^2+C}$$

$$\int \frac{dx}{(x^2+a^2)^{3/2}} = \frac{x/a^2}{\sqrt{x^2+a^2}} + C$$

$$\int \frac{xdx}{(x^2+a^2)^{3/2}} = -\frac{1}{\sqrt{x^2+a^2}} + C$$

$$\int \frac{x^2dx}{(x^2+a^2)^{3/2}} = \ln\left(\frac{\sqrt{x^2+a^2}}{a} + \frac{x}{a}\right) - \frac{x}{\sqrt{x^2+a^2}} + C$$

$$\int \frac{dx}{(x^2+a^2)^2} = \frac{1}{2a^2}\left(\frac{x}{x^2+a^2} + \frac{1}{a}\tan^{-1}\frac{x}{a}\right) + C$$

DEFINITE INTEGRALS

If m and n are integers, a, b, and c are constants,

$$\int_0^{\pi} \sin mx \sin nx dx = \int_0^{\pi} \cos mx \cos nx dx = \begin{cases} 0, & m \neq n \\ \pi/2, & m = n \end{cases}$$

$$\int_0^{\pi} \sin mx \cos nx dx = \begin{cases} 0, & m+n = \text{ even} \\ \dfrac{2m}{m^2 - n^2}, & m+n = \text{ odd} \end{cases}$$

$$\int_0^{2\pi} \sin mx \sin nx dx = \int_{-\pi}^{\pi} \sin mx \sin nx dx = \begin{cases} 0, & m \neq n \\ \pi, & m = n \end{cases}$$

$$\int_0^{\infty} \frac{\sin ax}{x} dx = \begin{cases} \pi/2, a > 0 \\ 0, a = 0 \\ -\pi/2, a < 0 \end{cases}$$

$$\int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$$

$$\int_0^{\infty} \frac{\sin^2 ax}{x^2} dx = |a| \frac{\pi}{2}, a > 0$$

In the following, $a > 0$ for the integrals to converge.

$$\int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\int_0^{\infty} e^{-ax^2} dx = \frac{1}{2}\sqrt{\frac{\pi}{a}}$$

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

$$\int_{-\infty}^{\infty} e^{-(ax^2+bx+c)} dx = \sqrt{\frac{\pi}{a}} e^{(bx^2-4ac)/4a}$$

$$\int_0^{\infty} e^{-ax} \cos b\, x dx = \frac{a}{a^2 + b^2}$$

$$\int_0^{\infty} e^{-ax} \sin b\, x dx = \frac{b}{a^2 + b^2}$$